Well-quasi-ordered classes of bounded clique-width main 63025d0c2bb13d7ba1f53e4ace38ba72f7888dc2 2026-01-08 10:38:40 +0100

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Abstract

We study classes of graphs with bounded clique-width that are well-quasi-ordered by the induced subgraph relation, in the presence of labels on the vertices. We prove that, given a finite presentation of a class of graphs, one can decide whether the class is labelled-well-quasi-ordered. This solves an open problem raised by Daligault, Rao and Thomassé in 2010. From our proof techniques, we also derive (restricted versions of) conjectures of Pouzet regarding well-quasi-ordering of graphs under the induced subgraph relation. Finally, we provide a structural characterization of those classes as those that are of bounded clique-width and do not existentially transduce the class of all finite paths.

CCS Concepts

• **Formal languages and automata theory**; • **Graph theory**;

Keywords

well-quasi-ordering, clique-width, automata theory, monoids, factorization forests, gap embedding

ACM Reference Format:
Aliaume Lopez and Maël Dumas. 2018. Well-quasi-ordered classes of bounded clique-width main 63025d0c2bb13d7ba1f53e4ace38ba72f7888dc2 2026-01-08 10:38:40 +0100 . *J. ACM* 37, 4, Article 111 (August 2018), 14 pages. https://doi.org/XXXXXXX.XXXXXXX

This document uses knowledge: a notion points to its *definition*.

1 Introduction

The theory of well-quasi-orderings (WQOs) provides a powerful combinatorial setting that has found applications in various areas of mathematics and computer science. In graph theory, the celebrated result of Robertson and Seymour [21] states that the class of all finite graphs is well-quasi-ordered under the minor relation, a profound result with deep algorithmic consequences. WQOs are also at the heart of *well-structured transition systems*, infinite state transitions systems over which verification algorithms can be designed [2, 3]. One of the appeal of WQOs is that they are closed under various operations, such as the sum and the product of WQOs. As an example, the closure of WQOs under the finite words, also known as Higman's lemma [15], has been used in the verification of so-called *lossy channel systems* [1].

Undirected finite graphs are naturally equipped with the *induced subgraph relation*, where G is an induced subgraph of H if G can be obtained from H by deleting vertices. Unlike the graph minor relation, the induced subgraph relation is not a well-quasi-ordering on the class of all finite graphs, as witnessed by the infinite family of cycles of increasing size, which form an infinite sequence of pairwise incomparable graphs. However, some classes of finite graphs are well-quasi-ordered under the induced subgraph relation: for instance, the class of all finite paths, or the class of all finite cliques. Distinguishing which classes of (finite) graphs are well-quasi-ordered under the induced subgraph relation is a long-standing open problem in graph theory, dating back to Pouzet's seminal work [19].

In general, there is little hope to characterise all such classes, since any countable order with finite infixes can be embedded into the induced subgraph relation on finite graphs (this is a consequence of [17, Lemma 5.1]). If one considers *labelled* well-quasi-orderings on classes of finite graphs, then there are high hopes of being able to provide a full characterisation of (labelled) well-quasi-ordered classes. Given a class C of finite graphs, and a well-quasi-ordering $(X, \leq)$, one can consider the class $\mathsf{Label}_X(C)$ of graphs in C whose vertices are labelled by elements of X, that is, equipped with a function $\ell\colon V(G) \to X$. The induced subgraph relation is then extended to labelled graphs by requiring that the labels are preserved by the embedding, i.e. that the label of a vertex in the smaller graph is less than or equal to the label of its image in the larger graph. A class C of finite graphs is said to be *$(X, \leq)$-well-quasi-ordered* if the class $\mathsf{Label}_X(C)$ is well-quasi-ordered under the labelled induced subgraph relation. We will say that it is *k-well-quasi-ordered* if it is $(X, \leq)$-well-quasi-ordered where X is a k-element antichain, and that it is *$\forall$-well-quasi-ordered* if it is $(X, \leq)$-well-quasi-ordered for every well-quasi-ordering $(X, \leq)$.

Example 1.1. The class Cycles of all finite cycles is not well-quasi-ordered, since the infinite sequence of cycles forms an infinite antichain. The class Paths of all finite paths is well-quasi-ordered since $(\mathsf{Paths}, \subseteq_i)$ is isomorphic to $(\mathbb{N}, \leq)$. However, it is not 2-well-quasi-ordered, since the infinite sequence of paths with colored endpoints forms an infinite antichain. The class Cliques of all finite cliques is $\forall$-well-quasi-ordered, since any X-labelled clique can be identified with a finite multiset of labels from X, and the set of

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ACM 1557-735X/2018/8-ART111
https://doi.org/XXXXXXX.XXXXXXX

Class	WQO	2-WQO	∀-WQO
Cycles	×	×	×
Paths	✓	×	×
Cliques	✓	✓	✓

Figure 1: Well-quasi-ordering properties of classes in Example 1.1. No example 2-WQO but not ∀-WQO is known.

finite multisets over a well-quasi-order is itself well-quasi-ordered by standard results on well-quasi-orders [10].

The study of labelled structures is a recurring theme in the theory of well-quasi-orderings: classical results such as Higman's lemma [15] or Kruskal's tree theorem [16] are actually stating that some classes of finite relational structures (respectively finite linear orders and finite meet trees) are well-quasi-ordered under the labelled embedding relation, for every well-quasi-ordering of labels. It was conjectured by Pouzet [19] that being ∀-well-quasi-ordered reduces to being 2-well-quasi-ordered for hereditary classes of structures (see [20, Problems 9, (1)] for a modern formulation). Another interesting conjecture of Pouzet [20, Problems 12] states that when a class C is ∀-well-quasi-ordered, one can add a total order on the vertices of its structures such that the new class is still ∀-well-quasi-ordered. This conjecture gained interest in the recent development of algebraic methods for polynomials over relational structures [14, Section 3, Theorem 11 and 12], where "monomials" are defined as structures labelled by $(\mathbb{N}, \leq)$. We refer to this property as being *orderably ∀-well-quasi-ordered*.

Even for classes of finite undirected graphs, these conjectures remain open. Several works dating back to the 90s have idendified structural properties of hereditary classes of finite graphs that guarantee that they are ∀-well-quasi-ordered, such as being of bounded tree-depth [13]. In 2010, Daligault, Rao and Thomassé [9] initiated the study in the reverse direction, by considering a syntax for graph classes (using NLC graph expressions), and identifying which ones among them are 2-well-quasi-ordered. In this context, they verified Pouzet's conjectures, but were unable to generalise their results to arbitrary graph subclasses [9, Conjecture 4]. All the classes considered in [9] have so-called *bounded clique-width*,[1] and it is believed that it is enough to understand hereditary classes of bounded clique-width to solve these conjectures in full generality, as stated in the following conjecture.

Conjecture 1.2 ([9, Conjecture 5]). *Let C be a hereditary class of finite graphs of that is 2-well-quasi-ordered. Then, C is of bounded clique-width.*

On parallel with the conjectures of Pouzet, another line of research has focused on finding structural obstructions to being ∀-well-quasi-ordered. The natural candidate for such an obstruction is the class of all finite paths Figure 1. For the particular classes studied in [9], it was shown that every class that is not 2-well-quasi-ordered contains all finite paths [9, Corollary 2]. There are examples of hereditary classes that are not 2-well-quasi-ordered and do not contain all finite paths, but on every known such example, one can "extract" finite paths using very simple logical formulas. This lead to the following conjecture, where existentially transduces is a notion from model theory (see Section 2 for a formal definition).

Conjecture 1.3 ([18, Conjecture 26]). *Let C be a hereditary class of finite graphs. If C is not 2-well-quasi-ordered, then it existentially transduces the class of all finite paths.*

Another kind of obstruction to being ∀-well-quasi-ordered comes from the existence of infinite antichains with a periodic structure that were called *periodic antichains* in [4, Section 7]. These antichains appear in [4, Conjecture 2], that considers about classes of graphs defined by finitely many excluded induced subgraphs, and relates the presence of these antichains to the fact that they are well-quasi-ordered. Since we are interested in 2-well-quasi-ordering, we propose hereafter an adaptation of this conjecture to our setting.

Conjecture 1.4 ([4, Adapted of Conjecture 2]). *Let C be a hereditary class of finite graphs that is not 2-well-quasi-ordered. Then, there exists an infinite (weak) periodic antichain in C.*

In 2024, Lopez [18] studied labelled well-quasi-orderings on classes of graphs of bounded *linear* clique-width, by leveraging combinatorial tools from automata theory (Simon's factorisation forests [22]). This allowed to prove a weakening of Pouzet's conjectures [18, Corollary 2], namely, that a hereditary class of bounded linear clique-width is ∀-well-quasi-ordered if and only if it is k-well-quasi-ordered for every $k \in \mathbb{N}$. It is apparent from the proof that the techniques also yield a natural ordering on the vertices whenever the class is ∀-well-quasi-ordered, although this is not formally stated in the paper. It was conjectured [18, Section 5] that these techniques could be refined to work on classes of bounded clique-width (not necessarily linear), and actually obtain the correspondence between 2-well-quasi-ordering and ∀-well-quasi-ordering.

Contributions. Our main contribution is Theorem 1.5, that answers positively to the conjectures of [18]. To that end, we first consider generalisation of the framework of [9] from NLC graph expressions to MSO-interpretations from trees (these will be formally defined in Section 2). This also generalises the setting of [18] that studied MSO-interpretations from *words*. Our main combinatorial result is the following.

Theorem 1.5. *Let C be the image of a class of finite trees under an MSO interpretation $\mathcal{I}$. Then, the following are equivalent and decidable given $\mathcal{I}$:*

(1) *C is 2-well-quasi-ordered,*
(2) *C is ∀-well-quasi-ordered,*
(3) *C is orderably ∀-well-quasi-ordered,*

From Theorem 1.5, we derive the following corollary that solves Pouzet's conjectures for hereditary classes of bounded clique-width.

Corollary 1.6. *The same holds when C is a hereditary class of bounded clique-width graphs.*

Because our proof techniques exhibit structural obstructions to being 2-well-quasi-ordered, we also obtain the following characterisations in Theorem 1.7. These answer positively to the conjectures of [4, 18]. In particular, for a hereditary class C to be 2-well-quasi-ordered, it suffices that it avoids very simple (weak) periodic antichains.

[1] This will be defined formally in Section 2

Theorem 1.7. *Let $\mathcal{C}$ be a hereditary class of finite graphs of bounded clique-width. Then, the following are equivalent:*

(1) *$\mathcal{C}$ is not 2-well-quasi-ordered,*
(2) *There exists a class $\mathcal{D} \subseteq \mathcal{C}$ of bounded linear clique-width which is not 2-well-quasi-ordered,*
(3) *There exists a strong periodic antichain in $\mathcal{C}$,*
(4) *There exists a weak periodic antichain in $\mathcal{C}$,*
(5) *$\mathcal{C}$ existentially transduces the class of all finite paths.*

We strongly believe that the results of Corollary 1.6 extend to non-hereditary classes of bounded clique-width graphs, but for the sake of clarity and space, we only state and prove them on hereditary ones. On the other hand, let us remark that Theorem 1.7 does not immediately extend to non-hereditary classes, as witnessed by the following example.

Example 1.8. The class $\mathcal{C}$ of complete binary trees has bounded clique-width, is not 2-well-quasi-ordered, but contains no periodic antichain, and every subclass $\mathcal{D} \subseteq \mathcal{C}$ of bounded linear clique-width is 2-well-quasi-ordered. Furthermore, $\mathcal{C}$ existentially transduces the class of all finite paths.

Proof Sketch. It is well-known that the class of complete binary trees has bounded clique-width. Furthermore, it is not 2-well-quasi-ordered, since one can colour the leaves and the root of the complete binary trees to form an infinite antichain. On the other hand, any subclass $\mathcal{D} \subseteq \mathcal{C}$ of bounded linear clique-width must be finite: indeed, complete binary trees have unbounded linear clique-width. Finally, because strong periodic antichains have bounded linear clique-width (see Section 6) there cannot be any strong periodic antichain in $\mathcal{C}$. The existential transduction of all finite paths from $\mathcal{C}$ is obtained by transducing all induced subgraphs (see Example 2.2), that in particular contains all finite paths. □

Outline. Section 2 introduces the necessary background on well-quasi-orders, graphs, MSO interpretations and first-order transductions. Section 3 shows how to reduce the study of graph classes obtained by MSO interpretations from trees to the study of trees labelled over finite monoids. This section also shows how to leverage forward Ramseyan splits to obtain well-behaved tree representations of graphs, that are well-quasi-ordered (as trees) with respect to the so-called *gap embedding* relation. Section 4 identifies combinatorial obstructions to being 2-well-quasi-ordered in images of interpretations from trees, and is the core combinatorial part of the paper. Section 5 brings together the previous results to prove our Theorem 1.5 and Corollary 1.6. We also discuss the equivalence between the first two items of Theorem 1.7. The remaining items of Theorem 1.7 are proven equivalent in Section 6. Finally, in Section 7 we discuss some perspectives and open problems.

2 Preliminaries

We assume the reader to be familiar with basic notions of logic on finite relational structures, such as monadic second-order logic (MSO), first-order logic and quantifier-free formulas.

Mael: TODO: unify (u, v) and uv)

Mael: Remarque globale : si tu dis qu'on ne parle que de graphe/arbre fini, alors pas besoin de repréciser fini à chaque fois.

Graphs. A *graph* is a pair $G = (V, E)$ where V is a set of *vertices* and $E \subseteq \binom{V}{2}$ is a set of *edges*. A graph is *undirected* if $(u, v) \in E$ implies $(v, u) \in E$ for all $u, v \in V$, and directed otherwise. In this paper we will focus on finite undirected graphs without self-loops. Given a set X, an *X-labelled graph* is a graph equipped with a function $\ell \colon V \to X$ that assigns a label to each vertex. Given a class $\mathcal{C}$ of finite undirected graphs, write $\mathsf{Label}_X(\mathcal{C})$ for the class of freely X-labelled graphs in $\mathcal{C}$, i.e., the class of all X-labelled graphs $G = (V, E, \ell)$ such that $G \in \mathcal{C}$.

Whenever $(X, \leq)$ is a quasi-order, we say that a graph G *embeds* into a graph H if there exists a *monomorphism* from G to H, i.e., an injective function $f \colon V_G \to V_H$ such that for all $u, v \in V_G$, $(u, v) \in E_G$ if and only if $(f(u), f(v)) \in E_H$, and such that for all $v \in V_G$, we have $\ell_G(v) \leq \ell_H(f(v))$. We write $G \subseteq_i H$ to denote that G embeds into H, which we also call the *induced subgraph relation*. When no order is specified along a set X, the order relation is assumed to be equality, i.e. $(X, =)$.

A class of graphs $\mathcal{C}$ is a *hereditary class* if it is closed under taking induced subgraphs, i.e. for $G \in \mathcal{C}$, if $H \subseteq_i G$, then $H \in \mathcal{C}$. The *hereditary closure* of a class of graphs $\mathcal{C}$ is the smallest hereditary class containing $\mathcal{C}$.

An *ordered graph* is a finite undirected graph $G = (V, E)$ equipped with a linear order $\leq_G$ on its vertices. The notion of induced subgraph relation extends to ordered graphs by requiring that the embedding $f \colon V_G \to V_H$ preserves the order, i.e., for all $u, v \in V_G$, $u \leq_G v$ if and only if $f(u) \leq_H f(v)$.

Well-quasi-orderings. A quasi-ordered set $(X, \leq)$ is a set X with a reflexive and transitive relation $\leq$. A sequence $(x_i)_{n \in \mathbb{N}}$ of elements in X is *good* if there exist $i < j$ such that $x_i \leq x_j$. A quasi-ordered set is *well-quasi-ordered* (*wqo*) if every infinite sequence is good. A *bad sequence* is an infinite sequence that is not good. A sequence of pairwise incomparable elements is called an *antichain*.

As mentioned in the introduction, several notions of well-quasi-ordering can be defined for a given class of finite undirected graphs $\mathcal{C}$. Let us briefly state them: $\mathcal{C}$ is *well-quasi-ordered* if the class $(\mathcal{C}, \subseteq_i)$ is well-quasi-ordered, $\mathcal{C}$ is *k-well-quasi-ordered* for some $k \in \mathbb{N}$, if for every set $(X, =)$ of size k, the class $(\mathsf{Label}_X(\mathcal{C}), \subseteq_i)$ is well-quasi-ordered, and $\mathcal{C}$ is *$\forall$-well-quasi-ordered* if for every well-quasi-order $(X, \leq)$, the class $(\mathsf{Label}_X(\mathcal{C}), \subseteq_i)$ is well-quasi-ordered.

Finally, a class of graph is *orderably well-quasi-ordered* (resp. *orderably k-well-quasi-ordered*, resp. *orderably $\forall$-well-quasi-ordered*) if there is a way to equip each graph G in $\mathcal{C}$ with a linear order $\leq_G$ on its vertices such that the resulting class of ordered graphs is well-quasi-ordered (resp. k-well-quasi-ordered, resp. $\forall$-well-quasi-ordered).

Trees. In this paper, *trees* are finite, binary, and ordered (have *left* and *right children*). They are understood as finite relational structures over the signature $\sigma = (\sqsubseteq, \preceq)$ where $\sqsubseteq$ is the *ancestor relation*, and $\preceq$ is the *sibling order* on the nodes of the tree, extended to all pairs of nodes as a topological order. In a tree $\mathcal{T}$, the *root* is denoted by $\mathsf{root}(\mathcal{T})$, with $\mathcal{T}$ being left implicit when clear from context. The set of *leaves* is denoted by $\mathsf{Leaves}(\mathcal{T})$. Given two nodes x and y in a tree T, their *least common ancestor* is denoted by $\mathsf{lca}(x, y)$. We will often use the *parent relation*, which state that y is the unique immediate ancestor of x.

As in the case of graphs, we will consider vertex-labelled trees where labels are taken from a well-quasi-order $(X, \leq)$. We will also consider edge-labelled trees, typically using a finite set of labels Σ. The class of vertex-labelled and edge-labelled trees is denoted by Trees_Σ^X. The sets Σ or X are omitted when unused (i.e., Trees denotes the class of unlabelled trees) Given two nodes x, y such that y is an *ancestor* of x, the notation $[x : y]_\mathcal{T}$ denotes the *word* in Σ^* obtained by reading the labels of the edges on the path from x to y in $\mathcal{T}$, we can leave $\mathcal{T}$ implicit if clear from context.

Bounded Clique-Width. An MSO-*interpretation* $\mathcal{I}$ from Trees_Σ to finite undirected graphs is given by a tuple of MSO formulas that specify how to respectively: select a subset of the leaves of tree to be the vertices of the graph $\phi_{\mathsf{univ}}(x)$, define the edges of the graph $\phi_{\mathsf{edge}}(x, y)$, and define the domain of the interpretation ϕ_{dom}. An interpretation $\mathcal{I}$ is called *simple* if $\phi_{\mathsf{univ}}(x)$ and ϕ_{dom} are tautologies.

Example 2.1. An MSO-interpretation $\mathcal{I}$ from Trees to finite undirected graphs that produces all paths can be defined as follows: ϕ_{dom} is the formula that is true on linear trees (i.e., trees where every node has at most one non-leaf child), $\phi_{\mathsf{univ}}(x)$ is the formula that is true if and only if x is a leaf and its sibling is not a leaf, $\phi_{\mathsf{edge}}(x, y)$ is the formula that is true if and only if are at distance exactly 3.

A class of graph has *bounded clique-width* if it is included in the image of some MSO-interpretation from finite trees to finite undirected graphs [8]. It has *bounded linear clique-width* if it is included in the image of some MSO-interpretation from linear trees to finite undirected graphs. Equivalently, a class of graphs has bounded linear clique-width if it is included in the image of some simple MSO-interpretation from finite words to finite undirected graphs.

Transductions from graphs to graphs. Another notion of logical transformation that will be relevant in this paper is that of existential transduction and existential interpretation that were mentioned in Theorem 1.7 and Conjecture 1.3, and are defined in a similar way as MSO-interpretations.

An *existential interpretation* from (labelled) graphs to graphs is given by an existential first order formula $\phi_{\mathsf{univ}}(x)$ that selects the vertices of the interpreted graph, an existential first order formula $\phi_{\mathsf{edge}}(x, y)$ that define the edges of the interpreted graph, and an existential first order formula ϕ_{dom} that selects the graphs of C on which the interpretation is defined. The semantics is that for every graph $G \in C$ satisfying ϕ_{dom}, the interpreted graph $\mathcal{I}(G)$ has for vertex set the vertices of G satisfying ϕ_{univ}, and has an edge between two vertices (u, v) if and only if G satisfies $\phi_{\mathsf{edge}}(u, v)$ (which is assumed to be symmetric and irreflexive on the selected vertices). A class C *existentially interprets* a class $\mathcal{D}$ if there exists an existential interpretation $\mathcal{I}$ from C to graphs such that $\mathcal{D} \subseteq \mathcal{I}(C)$. We say that C *existentially transduces* $\mathcal{D}$ if there exists a finite labelling set Σ such that $\mathsf{Label}_\Sigma(C)$ existentially interprets $\mathcal{D}$.

Example 2.2. Let C be a class of graphs. Then, C existentially transduces the hereditary closure of C, using labels $\Sigma = \{\circ, \bullet\}$, and the interpretation defined as follows: ϕ_{dom} is the tautological formula, $\phi_{\mathsf{univ}}(x)$ is the formula that is true if and only if x is labelled with $\bullet$, $\phi_{\mathsf{edge}}(x, y)$ is the formula that is true if and only if there is an edge between x and y in the original graph.

3 From Interpretations to Nested Trees

As a first step towards proving Theorem 1.5, we are going to transform a simple MSO-interpretation from trees to graphs into a more combinatorial object that will make the interpretation *quantifier-free*. To that end, we will chain two standard transformations from automata theory: the first one is to transform an MSO-interpretation into a so-called monoid interpretation, and the second one is to factorise the trees using forward Ramseyan splits (nested trees) [5]. Finally, we will show that the natural ordering on nested trees (the gap embedding) is *almost* such that the resulting interpretation is order preserving from nested trees to graphs.

Let us fix for the rest of the section a finite alphabet Σ and a simple MSO-interpretation $\mathcal{I}$ from Trees_Σ to finite undirected graphs defined by a formula $\varphi(x, y)$ that defines the edges of the graphs.

3.1 From Interpretation to Monoids

It follows from standard arguments that to a formula $\varphi(x, y)$ over Trees_Σ one can associate a finite monoid M and a morphism $\mu\colon \Sigma^* \to M$ such that for any tree T and any pair of leaves $x \preceq y$ in T, whether $T \models \varphi(x, y)$ is entirely determined by the values of $\mu([z : x]_\mathcal{T})$, $\mu([z : y]_\mathcal{T})$ and $\mu([\mathsf{root} : z]_\mathcal{T})$ where $z = \mathsf{lca}(x, y)$ is the least common ancestor of x and y. We refer to the book on Tree Automata Techniques and Applications for a comprehensive overview of the connection between Monadic Second Order Logic on trees, and tree automata [7]. In the rest of the section, we will fix such a monoid M and morphism μ. To simplify notations, we will write $[\![x : y]\!]_\mathcal{T}$ for $\mu([x : y]_\mathcal{T})$, that is, the product of the labels of the edges on the path from x to y in $\mathcal{T}$.

As a consequence, one can collect in a subset $P \subseteq M^3$ all the triples (a, b, c) corresponding to a satisfying assignments of the formula $\varphi(x, y)$. This is illustrated in Figure 2. We define a *monoid interpretation* from trees labelled over M to finite undirected graphs to be a tuple (Σ, μ, M, P) where Σ is a finite alphabet, $\mu\colon \Sigma^* \to M$ is a morphism to a finite monoid M, and $P \subseteq M^3$ is a subset of triples of elements of M. The interpretation works as follows: given a tree $\mathcal{T}$ whose edges are labelled by elements of Σ, the vertices of the interpreted graph are the leaves of T, and there is an edge between two leaves $x \preceq y$ if and only if the triple $([\![\mathsf{root} : \mathsf{lca}(x, y)]\!]_\mathcal{T}, [\![\mathsf{lca}(x, y) : x]\!]_\mathcal{T}, [\![\mathsf{lca}(x, y) : y]\!]_\mathcal{T})$ is in P.

Adding the function $[\![x : y]\!]_T$ to the signature of trees, and assuming that lca is also part of the signature, we can rewrite the formula $\varphi(x, y)$ as a quantifier-free formula as follows:

$$([\![\mathsf{root} : \mathsf{lca}(x, y)]\!]_\mathcal{T}, [\![\mathsf{lca}(x, y) : x]\!]_\mathcal{T}, [\![\mathsf{lca}(x, y) : y]\!]_\mathcal{T}) \in P \quad (1)$$

The main idea driving this paper is that understanding when a class of graphs is $\forall$-well-quasi-ordered can be reduced to finding a suitable well-quasi-order on trees such that the interpretation $\mathcal{I}$ is *order-preserving* from trees to graphs, leveraging the following standard Fact 3.1.

Fact 3.1 (Folklore). Let $(X, \leq_X)$ and $(Y, \leq_Y)$ be two quasi-orders, and $f\colon X \to Y$ be a surjective order-preserving map. If $(X, \leq_X)$ is well-quasi-ordered, then $(Y, \leq_Y)$ is well-quasi-ordered.

Similarly, if $f\colon X \to Y$ is an order reflection, that is, for every x_1, x_2 in X, $f(x_1) \leq_Y f(x_2)$ implies $x_1 \leq_X x_2$, and $(Y, \leq_Y)$ is well-quasi-ordered, then $(X, \leq_X)$ is well-quasi-ordered.

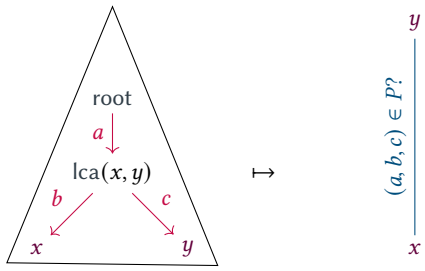

Figure 2: The interpretation of a tree using a monoid M and an accepting part $P \subseteq M^3$.

One candidate ordering on trees is the *composition ordering* defined as follows. Given two trees $\mathcal{T}_1$ and $\mathcal{T}_2$ labelled over a monoid M, we say that $\mathcal{T}_1$ is less than $\mathcal{T}_2$ in the composition ordering, written $\mathcal{T}_1 \preceq_{\mathsf{cmp}} \mathcal{T}_2$, if there exists a map $h\colon \mathcal{T}_1 \to \mathcal{T}_2$ that maps leaves to leaves, respects the relation lca, $\preceq$, and the function $[\![\cdot : \cdot]\!]$. This ordering is naturally extended to node-labelled trees by asking that for every node x in $\mathcal{T}_1$, the label of x is less or equal than the label of $h(x)$ in $\mathcal{T}_2$. It is straightforward to check that the interpretation $\mathcal{I}$ is order-preserving from trees equipped with the composition ordering to graphs equipped with the induced subgraph ordering. Hence, to prove that the image of $\mathcal{I}$ is $\forall$-well-quasi-ordered, it suffices to prove that the class of trees is well-quasi-ordered under the composition ordering when labelled using a well-quasi-ordered set $(X, \leq_X)$.

Historically, this approach was used to prove that some classes of graphs are labelled-well-quasi-ordered, see for instance the results of [9, 13]. Unfortunately, the composition ordering on trees is often way stricter than the induced subgraph ordering. It was observed that the composition ordering on trees is well-quasi-ordered if and only if *for every* $P \subseteq M^3$, the class of graphs obtained from trees by considering only the triples in P is labelled-well-quasi-ordered [18, Theorem 24]. To understand what happens for a precise choice of P, we need to have a finer understanding of the way trees can be decomposed.

3.2 Forward Ramseyan Splits

The combinatorial ingredient allowing us to further decompose the trees will come from an adaptation of the classical Simon Factorisation Theorem for semigroups [22], adapted to trees by Colcombet [5]. The rest of this section is devoted to explaining how this factorisation works, and how it can be used to define a more suitable ordering on trees.

A *split of height N* of a tree $\mathcal{T}$ is a mapping $\mathfrak{s}$ from the nodes of T to $\{1, \dots, N\}$. Given a split $\mathfrak{s}$ and two nodes $x \sqsubset_{\mathcal{T}} y$, we define $\mathfrak{s}(x\colon y)$ to be the minimal value of $\mathfrak{s}(z)$ for $x \sqsubset_{\mathcal{T}} z \sqsubset_{\mathcal{T}} y$, and $N+1$ otherwise. In particular, if y is a child of x, then $\mathfrak{s}(x\colon y) = N+1$.

Two elements $x \sqsubset_{\mathcal{T}} y$ such that $\mathfrak{s}(x) = \mathfrak{s}(y) = k$ are *k-neighbours* if $\mathfrak{s}(x\colon y) \geq k$. For every branch of the tree, being k-neighbours is an equivalence relation on the nodes of the tree having value k in the split belonging to said branch. A split induces a hierarchical structure on every branch of the tree, based on the k-neighbourhoods. We illustrate the situation in Figure 3. We will refer to the value of $\mathfrak{s}(x)$ as the *split depth* of the node x.

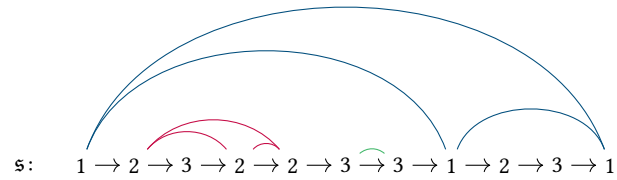

Figure 3: A split $\mathfrak{s}$ on a branch of a tree. The arcs in blue connect 1-neighbours, the arcs in red connect 2-neighbours, and the arcs in green connect 3-neighbours.

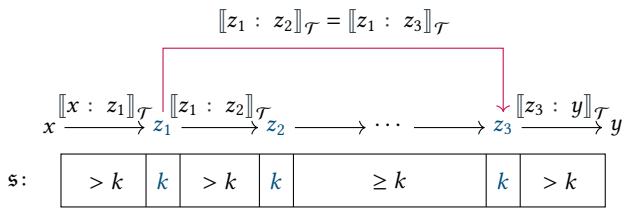

Figure 4: Fast computation of the value $[\![x : y]\!]_{\mathcal{T}}$ provided a forward Ramseyan split, using the fact that x and y are independent at level k.

A split $\mathfrak{s}$ is *forward Ramseyan* if for every $k \in \{1, \dots, N\}$, every branch b of the tree, and every x, y, x', y' in the same class of k-neighbourhood inside the branch b, with $x \sqsubset_{\mathcal{T}} y$ and $x' \sqsubset_{\mathcal{T}} y'$, we have:

$$[\![x : y]\!]_{\mathcal{T}} = [\![x : y]\!]_{\mathcal{T}} \cdot [\![x' : y']\!]_{\mathcal{T}} \quad . \tag{2}$$

In particular, $[\![x : y]\!]_{\mathcal{T}}$ is always an *idempotent*: $[\![x : y]\!]_{\mathcal{T}} \cdot [\![x : y]\!]_{\mathcal{T}} = [\![x : y]\!]_{\mathcal{T}}$. However, note that $[\![x : y]\!]_{\mathcal{T}}$ and $[\![x' : y']\!]_{\mathcal{T}}$ may be different idempotents.

Let us illustrate how one can use forward Ramseyan splits to compute efficiently the value of $[\![x : y]\!]_{\mathcal{T}}$ for any pair of nodes $x \sqsubset_{\mathcal{T}} y$. We say that x and y are *independent at level k* if $\mathfrak{s}(x : y) = k$ and there exist three nodes z_1, z_2, z_3 such that $x \sqsubseteq_{\mathcal{T}} z_1 \sqsubset_{\mathcal{T}} z_2 \sqsubseteq_{\mathcal{T}} z_3 \sqsubseteq_{\mathcal{T}} y$ and $\mathfrak{s}(z_1) = \mathfrak{s}(z_2) = \mathfrak{s}(z_3) = k$, and $\mathfrak{s}(x\colon z_1) > k$, $\mathfrak{s}(z_1\colon z_2) > k$, and $\mathfrak{s}(z_3\colon y) > k$. The idea being that x and y are *separated* by at least one k-neighbourhood. In the case of independence at level k, we can use the property of forward Ramseyan splits to compute $[\![x : y]\!]_{\mathcal{T}}$ without ever looking at the section between z_2 and z_3, leading to a fast (and first-order definable) computation. We refer to [5, Lemma 3] for more details on this computation.

The main theorem of [5] is that for every finite monoid M, there exists a finite depth N such that for every tree $\mathcal{T}$ edge-labelled using M, there exists a forward Ramseyan split of height N for $\mathcal{T}$. We can therefore assume that our trees are always equipped with a forward Ramseyan split of height N.

Let us now state the main remark that will guide the rest of this paper: given a branch of a tree equipped with a forward Ramseyan split as the one illustrated in Figure 4, the value of $[\![x : y]\!]_{\mathcal{T}}$ does not change when inserting new nodes in the section between z_2 and z_3 as long as these new nodes are assigned a value of the split bigger than or equal to k, and the resulting tree is still forward Ramseyan. A fortiori, whether there is an edge between two leaves of x and y

in the resulting graph is not influenced by such insertions. This will guide our definition of a suitable ordering on trees in the upcoming Section 3.3.

3.3 Marked Nested Trees

In this section, we will show how the notion of forward Ramseyan split gets us closer to defining a well-quasi-order on trees that is suitable for our purpose. A first step is to notice that there already exists a notion of embedding that somehow respects forward Ramseyan splits, called gap-embeddings and that endows trees with a well-quasi-order [11].

Definition 3.2 ([11, Definition 3.3]). A *gap-embedding* between trees $\mathcal{T}_1, \mathcal{T}_2$ endowed with forward Ramseyan splits $\mathfrak{s}_1, \mathfrak{s}_2$ is a mapping $h\colon \mathcal{T}_1 \to \mathcal{T}_2$ that

(1) is a tree embedding: respects the ancestor relation $\sqsubseteq$, least common ancestors lca, and sibling ordering $\preceq$,
(2) satisfies the root gap property: $\mathfrak{s}_2(r_2 : h(r_1)) \geq \mathfrak{s}_1(r_1)$ where r_1, r_2 are the roots of $\mathcal{T}_1, \mathcal{T}_2$,
(3) satisfies the edge gap property: for every edge $x \sqsubset_{\mathcal{T}} y$ in $\mathcal{T}_1$, $\mathfrak{s}_2(h(x) : h(y)) \geq \mathfrak{s}_1(y)$.

If the trees are node-labelled by a set $(X, \leq_X)$, we further require that for every node x in $\mathcal{T}_1$, the label of x is less than or equal to the label of $h(x)$ in $\mathcal{T}_2$.

In the following Fact 3.3 we illustrate why gap-embeddings are well-suited for our purpose: they respect the gaps induced by forward Ramseyan splits.

Fact 3.3. Let $\mathcal{T}_1, \mathcal{T}_2$ be two trees and $\mathfrak{s}_1, \mathfrak{s}_2$ be two forward Ramseyan splits of height N of $\mathcal{T}_1, \mathcal{T}_2$ Assume that $h\colon \mathcal{T}_1 \to \mathcal{T}_2$ is a gap-embedding from $(\mathcal{T}_1, \mathfrak{s}_1)$ to $(\mathcal{T}_2, \mathfrak{s}_2)$. Then, for every u, v in $\mathcal{T}_1$, if $\mathfrak{s}_1(u : v) > k$ and $\mathfrak{s}_1(v) = k$ for some $k \in \{1, \dots, N\}$, then $\mathfrak{s}_2(h(u) : h(v)) \geq k$.

Proof. This is proven by induction on the length of the path from u to v in $\mathcal{T}_1$. The base case is exactly Item 3 of the definition of gap-embedding. □

One other key property of the gap-embedding ordering is that it is a well-quasi-ordering.

Theorem 3.4 ([11, Theorem 3.1, Main Theorem]). *The class of trees equipped with forward Ramseyan splits is well-quasi-ordered under the gap-embedding ordering, even under the assumption that nodes are labelled with a well-quasi-ordered set $(X, \leq_X)$.*

We will now devise a suitable modification of the gap-embedding ordering that will ensure that the interpretation $\mathcal{I}$ is order-preserving from trees to graphs. In particular, this modification will take into account the edge-labelling of the trees using the monoid M, that are not considered in the gap-embedding ordering.

Definition 3.5. A *marked nested tree* is a tuple $(\mathcal{T}, \mathfrak{s}, \rho)$ where $\mathcal{T}$ is a tree whose edges are labelled over a finite monoid M, $\mathfrak{s}$ is a forward Ramseyan split of height N of $\mathcal{T}$, and ρ is a function from the nodes of $\mathcal{T}$ to $\{\mathbf{M}, \mathbf{S}, \mathbf{D}\}$, respectively called *marked*, *separating*, and *dummy* nodes.

A marked nested tree $(\mathcal{T}, \mathfrak{s}, \rho)$ is *well-marked* if the marked nodes contain the root of $\mathcal{T}$, are closed under the least common ancestor operation, and furthermore satisfy that for every $x \sqsubset_{\mathcal{T}} z_1 \sqsubset_{\mathcal{T}} y$ such that $\mathfrak{s}(x : z_1) > k$, $\mathfrak{s}(z_1) = k$, and $\mathfrak{s}(z_1 : y) \geq k$ for some $k \in \{1, \dots, N\}$, if x and y are marked nodes, then there exists z_2, z_3 such that $x \sqsubset_{\mathcal{T}} z_1 \sqsubset_{\mathcal{T}} z_2 \sqsubseteq_{\mathcal{T}} z_3 \sqsubseteq_{\mathcal{T}} y$, $\mathfrak{s}(z_1) = \mathfrak{s}(z_2) = \mathfrak{s}(z_3) = k$, $\mathfrak{s}(x : z_1) > k$, $\mathfrak{s}(z_1 : z_2) > k$, $\mathfrak{s}(z_2 : z_3) \geq k$, and $\mathfrak{s}(z_3 : y) \geq k$, with $\rho(z_1) = \mathbf{M}$, and $\rho(z_2) \in \{\mathbf{M}, \mathbf{S}\}$. A graphical illustration of the condition is provided below:

$$\begin{tikzcd}
\overset{\mathbf{M}}{x} \arrow[r, "> k"'] & \overset{\mathbf{M}}{z_1} \arrow[r, "> k"', "k"] & \overset{\in \{\mathbf{M},\mathbf{S}\}}{z_2} \arrow[r, "\geq k"', "k"] & \overset{\in \{\mathbf{M},\mathbf{S},\mathbf{D}\}}{z_3} \arrow[r, "\geq k"', "k"] & \overset{\mathbf{M}}{y}
\end{tikzcd}$$

Definition 3.6. Given two marked nested trees $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1)$ and $(\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$, we say that $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1)$ *gap-embeds* into $(\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$, written $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1) \leq_{\text{gap}} (\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$, if there exists a mapping $h\colon \mathcal{T}_1 \to \mathcal{T}_2$ that

★ is a gap embedding from $(\mathcal{T}_1, \mathfrak{s}_1)$ to $(\mathcal{T}_2, \mathfrak{s}_2)$,
(4) sends the root of $\mathcal{T}_1$ to the root of $\mathcal{T}_2$,
(5) maps leaves to leaves,
(6) respects the marking: $\rho_1 = \rho_2 \circ h$,
(7) respects local products: if y is the left (resp. right) child of x in $\mathcal{T}_1$, and y' is the left (resp. right) child of $h(x)$ in $\mathcal{T}_2$, then $[\![x : y]\!]_{\mathcal{T}_1} = [\![h(x) : y']\!]_{\mathcal{T}_2}$.
(8) respects neighbourhood products: for every $k \in \{1, \dots, N\}$, and every node $x \in \mathcal{T}_1$, if z is the closest ancestor of x such that $\mathfrak{s}_1(z) = k$ and z' is the closest ancestor of $h(x)$ such that $\mathfrak{s}_2(z') = k$, then $[\![z : x]\!]_{\mathcal{T}_1} = [\![z' : h(x)]\!]_{\mathcal{T}_2}$.
(9) is gluing on non-dummy nodes: if $\rho_1(y) \in \{\mathbf{M}, \mathbf{S}\}$, and x is the parent of y in $\mathcal{T}_1$, then $h(x)$ is the parent of $h(y)$ in $\mathcal{T}_2$.

Unlike the gap-embedding ordering of Theorem 3.4, the marked gap-embedding ordering of Definition 3.6 is not a well-quasi-order because of the gluing condition of Item 9. We can create an infinite antichain by considering the infinite sequence of trees $(\mathcal{T}_n, \mathfrak{s}_n, \rho_n)$ where $\mathcal{T}_n$ is a branch of length $n + 1$, $\mathfrak{s}_n$ assigns the value 1 to every node of $\mathcal{T}_n$, and ρ_n marks all the nodes of $\mathcal{T}_n$ with $\mathbf{M}$. It is straightforward to check that for every $n < m$, $(\mathcal{T}_n, \mathfrak{s}_n, \rho_n)$ does not gap-embed into $(\mathcal{T}_m, \mathfrak{s}_m, \rho_m)$.

To recover the well-quasi-order property, we will restrict our attention to trees that make a limited use of the gluing property. A L-bounded marked nested tree is a marked nested tree such that any path of non dummy nodes has length at most L. As we will see in the following Theorem 3.7, the class of L-bounded well-marked nested trees is well-quasi-ordered under the marked gap-embedding ordering when labelled over a well-quasi-ordered set.

Theorem 3.7. *For every finite L, the class of L-bounded well-marked nested trees labelled over a well-quasi-ordered set $(X, \leq_X)$ is well-quasi-ordered under the gap-embedding ordering.*

Proof. We provide a function f that maps L-bounded well-marked nested trees into nested trees labelled over a well-quasi-ordered set, such that if there exists a gap-embedding between $f(\mathcal{T}_1, \mathfrak{s}_1, \rho_1)$ and $f(\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$, then $\mathcal{T}_1 \leq_{\text{gap}} \mathcal{T}_2$ as marked nested trees. Using Theorem 3.4, this will conclude the proof.

One can encode Items 4 to 8 of Definition 3.6 by adding suitable labels to the nodes of the trees. Formally, we define a new labelling of the nodes of a tree $\mathcal{T}$ as follows: we add to each node x of $\mathcal{T}$

the label $[\![z : x]\!]_{\mathcal{T}}$ for every $k \in \{1, \dots, N\}$, where z is the least ancestor of x such that $\mathfrak{s}(z) = k$, or a special symbol $\perp$ if no such ancestor exists. Furthermore, if y is the immediate left (resp. right) child of x in $\mathcal{T}$, we also add the label $[\![x : y]\!]_{\mathcal{T}}$ to x, or the special symbol $\perp$ if no such child exists. We also distinguish the root by adding a special label root, and we add a special label leaf to every leaf. Finally, we also add the label $\rho(x)$ to x. Since M is finite, the new labelling is still over a well-quasi-ordered set.

It remains to ensure that the mapping h is gluing on non-dummy nodes. To that end, we will modify the split $\mathfrak{s}$ into a new split $\mathfrak{s}'$ as follows: for every non-dummy node x, we have that x is the ith element in a maximal path of non-dummy nodes $x_1 \sqsubset_{\mathcal{T}} x_2 \sqsubset_{\mathcal{T}} \cdots \sqsubset_{\mathcal{T}} x_n$ that contains x. Because the tree is L-bounded, we have $n \leq L$. We set $\mathfrak{s}'(x) = N + 1 + i$. For a dummy node y, we set $\mathfrak{s}'(y) = \mathfrak{s}(y)$.

Note that any gap-embedding h from $(\mathcal{T}_1, \mathfrak{s}'_1)$ to $(\mathcal{T}_2, \mathfrak{s}'_2)$ is necessarily gluing on non-dummy nodes: assume that $\rho_1(y) \in \{\mathbf{M}, \mathbf{S}\}$, and let x be the parent of y in $\mathcal{T}_1$. Since $\mathfrak{s}'_1(y) = N + 1 + i$, we know that $\mathfrak{s}'_2(h(x) \colon h(y)) \geq \mathfrak{s}'_1(y) = N + 1 + i$. But this means that between $h(x)$ and $h(y)$ there are only nodes with split depth greater or equal than $N + 1 + i$. If $i = 1$ (i.e., y is the first node of a maximal path of non-dummy nodes), then there is no such node, and $h(x)$ is the parent of $h(y)$. Otherwise, $\mathfrak{s}'_1(x) = N + i > N$, and therefore the path from $h(x)$ to $h(y)$ contains only non dummy nodes. Since the split values have been chosen to be strictly increasing inside maximal paths of non-dummy nodes, there is no such node, and $h(x)$ is the parent of $h(y)$. □

Lemma 3.8. *Let $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1)$ and $(\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$ be two marked nested trees such that $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1) \leq_{\text{gap}} (\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$ via some mapping h. For every pair of marked nodes $x \sqsubset_{\mathcal{T}_1} y$ in $\mathcal{T}_1$, we have $[\![x : y]\!]_{\mathcal{T}_1} = [\![h(x) : h(y)]\!]_{\mathcal{T}_2}$.*

Proof. We proceed by induction on the value of $\mathfrak{s}_1(x \colon y)$. If $\mathfrak{s}_1(x \colon y) = N + 1$, then y is a child of x. Because y is a marked node, then by Item 9 we know that $h(y)$ is a child of $h(x)$. Therefore, by Item 7, we have $[\![x : y]\!]_{\mathcal{T}_1} = [\![h(x) : h(y)]\!]_{\mathcal{T}_2}$.

Otherwise, let $k = \mathfrak{s}_1(x : y) \leq N$. Since $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1)$ is well-marked, there exists z_1, z_2, z_3 such that $x \sqsubset_{\mathcal{T}_1} z_1 \sqsubset_{\mathcal{T}_1} z_2 \sqsubseteq_{\mathcal{T}_1} z_3 \sqsubseteq_{\mathcal{T}_1} y$, $\mathfrak{s}_1(z_1) = \mathfrak{s}_1(z_2) = \mathfrak{s}_1(z_3) = k$, and $\mathfrak{s}_1(x \colon z_1) > k$, $\mathfrak{s}_1(z_1 \colon z_2) > k$, $\mathfrak{s}_1(z_2 \colon z_3) \geq k$, and $\mathfrak{s}_1(z_3 \colon y) > k$, with $\rho_1(z_1) = \mathbf{M}$, and $\rho_1(z_2) \in \{\mathbf{M}, \mathbf{S}\}$. This is depicted in the following figure:

$$x \xrightarrow[>k]{[\![x : z_1]\!]_{\mathcal{T}_1}} \underset{k}{z_1} \xrightarrow[>k]{[\![z_1 : z_2]\!]_{\mathcal{T}_1}} \underset{k}{z_2} \xrightarrow[\geq k]{[\![z_2 : z_3]\!]_{\mathcal{T}_1}} \underset{k}{z_3} \xrightarrow[>k]{[\![z_3 : y]\!]_{\mathcal{T}_1}} y$$

By applying the gap-embedding h, we have the following configuration inside $\mathcal{T}_2$, where we used the notations $a = [\![h(x) : h(z_1)]\!]_{\mathcal{T}_2}$, $b = [\![h(z_1) : h(z_2)]\!]_{\mathcal{T}_2}$, $c = [\![h(z_2) : h(z_3)]\!]_{\mathcal{T}_2}$, and $d = [\![h(z_3) : h(y)]\!]_{\mathcal{T}_2}$.

$$h(x) \xrightarrow[>k]{a} \underset{k}{h(z_1)} \xrightarrow[>k]{b} \underset{k}{h(z_2)} \xrightarrow[\geq k]{c} \underset{k}{h(z_3)} \xrightarrow[>k]{d} h(y)$$

Let us briefly argue the inequalities on the splits on the above paths. The non-strict inequalities follow from Fact 3.3. Furthermore, if v is a marked node or a separating node, then we can strengthen the inequality to a strict one, i.e., $\mathfrak{s}(h(u) \colon h(v)) > k$ even when $\mathfrak{s}(v) \neq k$. This is because of Item 9, which ensures that the immediate parent of $h(v)$ is the image of the immediate parent of v, which has a split value strictly greater than k, and we conclude by the previous argument.[2]

Using these facts we conclude that:

$$\begin{aligned}
\mathfrak{s}_2(h(x) \colon h(z_1)) &> k && \text{because } z_1 \text{ is a marked,} \\
\mathfrak{s}_2(h(z_1) \colon h(z_2)) &> k && \text{because } z_2 \text{ is a marked or separating,} \\
\mathfrak{s}_2(h(z_2) \colon h(z_3)) &\geq k && \text{by Fact 3.3,} \\
\mathfrak{s}_2(h(z_3) \colon h(y)) &> k && \text{because } y \text{ is a marked,}
\end{aligned}$$

Now, we can compute the values of a, b, and d:

$$\begin{aligned}
[\![x : z_1]\!]_{\mathcal{T}_1} &= a && \text{by induction hypothesis,} \\
[\![z_1 : z_2]\!]_{\mathcal{T}_1} &= b && \text{by Item 8,} \\
[\![z_3 : y]\!]_{\mathcal{T}_1} &= d && \text{by Item 8.}
\end{aligned}$$

Finally, because $h(x)$ and $h(y)$ are independent at level k, and so are x and y, we have by Equation (2) that

$$\begin{aligned}
[\![x : y]\!]_{\mathcal{T}_1} &= [\![x : z_1]\!]_{\mathcal{T}_1} \cdot [\![z_1 : z_2]\!]_{\mathcal{T}_1} \cdot [\![z_2 : z_3]\!]_{\mathcal{T}_1} \cdot [\![z_3 : y]\!]_{\mathcal{T}_1} \\
&= [\![x : z_1]\!]_{\mathcal{T}_1} \cdot [\![z_1 : z_2]\!]_{\mathcal{T}_1} \cdot [\![z_3 : y]\!]_{\mathcal{T}_1} \\
&= a \cdot b \cdot d \\
&= a \cdot b \cdot c \cdot d \\
&= [\![h(x) : h(y)]\!]_{\mathcal{T}_2} \quad . \qquad \square
\end{aligned}$$

Corollary 3.9. *Let $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1)$ and $(\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$ be two well-marked nested trees such that $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1) \leq_{\text{gap}} (\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$. Then, the graph interpreted from $(\mathcal{T}_1, \mathfrak{s}_1, \rho_1)$ is an induced subgraph of the graph interpreted from $(\mathcal{T}_2, \mathfrak{s}_2, \rho_2)$ when considering only marked leaves as vertices.*

Proof. Follows directly from Lemma 3.8 and the definition of the monoid interpretation using the set $P \subseteq M^3$ (see Equation (1)): edges between two marked leaves x and y only depend on the values of $[\![\text{root} : \text{lca}(x, y)]\!]_{\mathcal{T}}$, $[\![\text{lca}(x, y) : x]\!]_{\mathcal{T}}$, and $[\![\text{lca}(x, y) : y]\!]_{\mathcal{T}}$, all of which are marked nodes. □

4 Bad Patterns in Images of Interpretations

In this section, we fix a monoid interpretation $\mathcal{I} = (\Sigma, \mu, M, P)$ from trees edge-labelled over a finite alphabet Σ to finite undirected graphs. We furthermore assume that all trees we consider are equipped with a forward Ramseyan split $\mathfrak{s}$ over the monoid M. Our goal is to isolate some combinatorial obstructions to being 2-well-quasi-ordered in classes of graphs that are images of monoid interpretations from trees. The key idea of this section is to start from a tree $\mathcal{T}$ equipped with a forward Ramseyan split $\mathfrak{s}$, and try to understand under which conditions one can turn this tree into a L-bounded marked nested tree representing the same graph, allowing us to leverage Theorem 3.7 to conclude that the class of graphs is $\forall$-well-quasi-ordered.

[2]The case where there are no intermediate nodes between u and v happens if v is the immediate child of u, in which case the property is given directly by the gluing property.

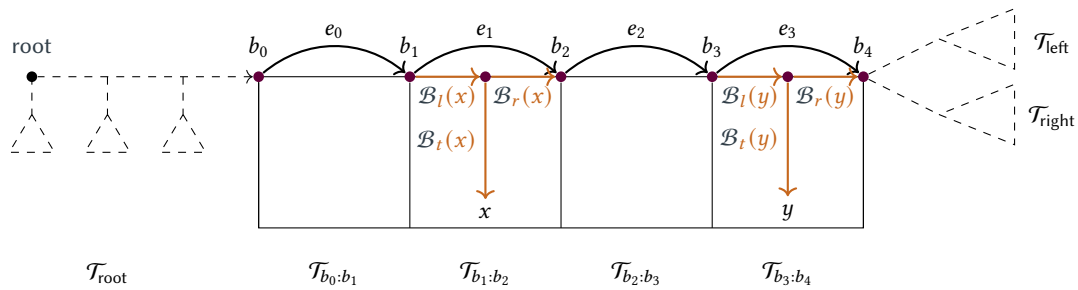

Figure 5: Partitioning a branch of a tree $\mathcal{T}$ using a bough B. Here, the nodes b_i for $0 \leq i \leq 4$ are k-neighbours forming the backbone B. The letter e_i denotes idempotent monoid element $[\![b_i : b_{i+1}]\!]_{\mathcal{T}}$. In dashed, we represented the context $C[\square]$ such that $\mathcal{T} = C[B]$. To compute the presence of an edge between x and y in the resulting graph, it is sufficient to know the values of $[\![b_2 : b_3]\!]_{\mathcal{T}}$, and the respective bough types of x and y in their respective bough blocks $\mathcal{T}_{b_1:b_2}$ and $\mathcal{T}_{b_3:b_4}$.

Definition 4.1. Let $\mathcal{T}$ be a tree and $\mathfrak{s}$ be a forward Ramseyan split of height N. Let $n \geq 1$, and let $b_0, b_1, \ldots, b_n$ nodes on branch of $\mathcal{T}$ such that $b_0 \sqsubseteq b_1 \sqsubseteq \cdots \sqsubseteq b_n$, and for every $0 \leq i < n$, $\mathfrak{s}(b_i) = k$, and $\mathfrak{s}(b_i : b_{i+1}) > k$. A *bough of level k* is the subtree of $\mathcal{T}$ induced by all nodes x such that $b_0 \sqsubseteq x$, and $\neg(b_n \sqsubseteq x)$, plus b_n itself.

We say that the *dimension* of the bough B is n, and that $b_0, b_1, \ldots, b_n$ form the *backbone* of B.

Let B be a bough of level k in $\mathcal{T}$ and let $b_0, b_1, \ldots, b_n$ be its backbone. Given indices $i < j$, we define the $\mathcal{T}_{b_i:b_j}$ as the set of nodes x in $\mathcal{T}$ such that $b_i \sqsubseteq x$ and $\neg(b_j \sqsubseteq x)$. When $j = i + 1$, this set of nodes is called a *block*. We also refer to Figure 5 for an illustration of the resulting partition of the tree $\mathcal{T}$ with respect to a given bough B.

For every leaf $x \in \mathcal{T}_{b_0:b_n}$, we define $\mathsf{B}_t(x) \triangleq \mathsf{lca}(x, b_n)$. Moreover, we define $\mathsf{B}_l(x)$ (resp. $\mathsf{B}_r(x)$) to be the maximal (resp. minimal) element of the backbone $b_0, b_1, \ldots, b_n$ such that $\mathsf{B}_l(x) \sqsubseteq \mathsf{B}_t(x) \sqsubseteq \mathsf{B}_r(x)$. Using these reference points, we define the following monoid elements associated to a leaf x in a block of B:

$$\begin{aligned}
\mathcal{B}_t(x) &\triangleq [\![\mathsf{B}_t(x) : x]\!]_{\mathcal{T}}, \\
\mathcal{B}_l(x) &\triangleq [\![\mathsf{B}_l(x) : \mathsf{B}_t(x)]\!]_{\mathcal{T}}, \\
\mathcal{B}_r(x) &\triangleq [\![\mathsf{B}_t(x) : \mathsf{B}_r(x)]\!]_{\mathcal{T}}, \\
\mathcal{B}_{\text{root}}(x) &\triangleq [\![\text{root} : \mathsf{B}_l(x)]\!]_{\mathcal{T}}.
\end{aligned}$$

We call this tuple the *bough type* of the leaf x with respect to the bough B. We again refer to Figure 5 for an illustration of the type of a leaf with respect to a given bough B. There are only finitely many possible bough types because M is finite.

In the rest of the paper, we will often rely on some tree surgery operation consisting in replacing a bough in a tree by some other bough. Given a bough B in a tree $\mathcal{T}$ defined using a backbone b_0, $b_1, \ldots, b_n$, we write $\mathcal{T} = C[B]$ to denote that $\mathcal{T}$ can be obtained by plugging the bough B into a *context* $C[\square]$. Formally, a *context* $C[\square]$ is given by a tree $\mathcal{T}_{\text{root}}$ with a distinguished leaf $\square$, together with two trees $\mathcal{T}_{\text{left}}$ and $\mathcal{T}_{\text{right}}$, and two monoid elements m_{left} and m_{right}. These three elements are represented as dashed lines in Figure 5. The tree $C[B]$ is then obtained as follows: one first attaches $\mathcal{T}_{\text{left}}$ as the left subtree of b_n with an edge labelled m_{left}, then one attaches $\mathcal{T}_{\text{right}}$ as the right subtree of b_n with an edge labelled m_{right}, and finally, one replaces the distinguished leaf $\square$ in $\mathcal{T}_{\text{root}}$ by the node b_0.

In order to simplify statements, we will say that the *leaves* of a context $C[\square]$ are the leaves of the tree $\mathcal{T}_{\text{root}}$, $\mathcal{T}_{\text{left}}$, and $\mathcal{T}_{\text{right}}$, excluding the distinguished leaf $\square$. As a consequence, the leaves of $C[B]$ are exactly the leaves of $C[\square]$ together with the leaves of the blocks of B. Furthermore, we will often need to talk about the *first idempotent value* of a bough B defined using the backbone $b_0, b_1, \ldots, b_n$, which is the idempotent element $[\![b_0 : b_1]\!]_{\mathcal{T}}$.

Given two boughs B and B' of level k, and a context $C[\square]$, one can consider the two trees $C[B]$ and $C[B']$. These two trees have edges labelled with M, and are equipped with splits induced by the splits on $C[\square]$ and B (resp. B'). We say that B and B' are *compatible* if they share the same first idempotent value, and if for every context $C[\square]$ such that the tree $C[B]$ (with split $\mathfrak{s}$) is forward Ramseyan, the tree $C[B']$ (with split $\mathfrak{s}'$) is also forward Ramseyan.

Definition 4.2. Let B be a bough of level k in a tree $\mathcal{T} = C[B]$. We say that B is a *good bough* if, there exists a compatible bough H of level k, and a map $h \colon \mathcal{I}(C[B]) \to \mathcal{I}(C[H])$ such that:

(1) h is an embedding of graphs,
(2) h is the identity map on leaves of $C[\square]$,
(3) there exist three consecutive blocks in H such that all their leaves are left untouched by h.

Mael: whose leaves are left untouched by h.

A *bad bough* is a bough that is not a good bough.

We claim that the existence of bad boughs of arbitrarily large dimension is the only obstruction to being 2-well-quasi-ordered.

Lemma 4.3. *The following are equivalent:*

(1) *The image of $\mathcal{I}$ is 2-well-quasi-ordered.*
(2) *The image of $\mathcal{I}$ is $\forall$-well-quasi-ordered.*
(3) *There exists a bound on the dimension of bad boughs.*

It is clear that Item 2 implies Item 1. In Section 4.1 we will prove that Item 3 implies Item 2, while in Section 4.2 we will prove that Item 1 implies Item 3, concluding the proof of Lemma 4.3.

4.1 Using the Gap Embedding

In this section, we will leverage Theorem 3.7 to prove that Item 3 implies Item 2 of Lemma 4.3. The main idea will be to use the bound on the dimension of bad boughs to insert "dummy nodes" inside

our trees, turning them into marked nested trees. Because we can insert these dummy nodes in a controlled manner, we will be able to ensure that the resulting marked nested trees are L-bounded for some $L \in \mathbb{N}$, allowing us to apply Theorem 3.7. This is exactly the content of the following Lemma 4.4.

Lemma 4.4. *Assume that there exists a bound N_0 on the dimension of bad boughs. Then, for every tree $\mathcal{T}$ equipped with a forward Ramseyan split $\mathfrak{s}$, one can effectively construct a well-marked nested tree $(\mathcal{T}', \mathfrak{s}', \rho)$ that is L-bounded for some $L \in \mathbb{N}$, and such that $\mathcal{I}(\mathcal{T})$ is an induced subgraph of $\mathcal{I}(\mathcal{T}')$, when the latter is restricted to vertices corresponding to marked leaves.*

Proof. We proceed iteratively starting from the tree $\mathcal{T}$ where every node is set as a marked node. A rewrite step is as follows: given a bough backbone $b_0, b_1, \dots, b_n$ defining a bough B of level k, where all nodes of the bough backbone are marked nodes, and $n > N_0$. We know that B is a good bough by assumption and write $\mathcal{T} = C[B]$.

By definition of good boughs, there exists a compatible bough H and a map $h\colon \mathcal{I}(C[B]) \to \mathcal{I}(C[H])$ satisfying the three conditions of Definition 4.2. Let us define $\mathcal{T}' = C[H]$, which is still forward Ramseyan by definition of compatible boughs. Let us now define the marking of $\mathcal{T}'$. For nodes of $C[\square]$, we keep the same marking as in $\mathcal{T}$. For nodes of H, we identify the three consecutive blocks whose leaves are left untouched by h as $H_{z_i:z_{i+1}}$, $H_{z_{i+1}:z_{i+2}}$, and $H_{z_{i+2}:z_{i+3}}$. We set all of these nodes as dummy nodes except z_i and z_{i+3} that are set as marked nodes, and z_{i+1} that is set as a separating node. Finally, all other nodes of H are set as marked nodes.

We claim that one can apply these rewrite steps as follows: first, apply them to boughs of minimal split level until exhaustion, and then proceed with the next split level, and so on. To a tree $\mathcal{T}$ and a split level k, one can associate the multiset of dimensions of all boughs of level k with backbone made of marked nodes inside $\mathcal{T}$. Since each rewrite step decreases this multiset for the Dershowitz–Manna ordering this process terminates [12].

Let us first observe that every rewrite step preserves the fact that $\mathcal{T}$ is a well marked nested tree. This is because non-marked nodes are only inserted after a separating node.

Let us now prove that every rewrite step preserves the fact that the graph $\mathcal{I}(\mathcal{T})$ restricted to vertices corresponding to marked leaves is an induced subgraph of $\mathcal{I}(\mathcal{T}')$ restricted to vertices corresponding to marked leaves. This is a direct consequence of the fact that $h\colon \mathcal{I}(C[B]) \to \mathcal{I}(C[H])$ is an embedding of graphs, and of the fact that h is the identity on leaves of $C[\square]$.

From our construction, it is clear that at every level k, there are at most $N_0 + 1$ consecutive marked nodes in the bough backbone of any bough of level k.

Mael: I think I have a problem here with bough backbone. Say you have B with $b_0, ..., b_n$. After the surgery, $h(b_0), ..., h(b_n)$ are still marked and are also all in the same k-neighbourhood. I think you must specify that you cannot "expand" by adding another b in th sequence, you need some maximality notion.

Let us now prove that the resulting tree $\mathcal{T}'$ is L-bounded for some $L \in \mathbb{N}$. Assume towards a contradiction that there exists arbitrarily long paths of non-dummy nodes in the resulting tree $\mathcal{T}'$. To a pair of nodes $x \sqsubseteq y$ in such a path, we associate the colour $(\mathfrak{s}'(x), \mathfrak{s}'(x\colon y), \mathfrak{s}'(y))$. By a Ramsey argument, if the path is too long, there exists a sequence $x_0 \sqsubseteq x_1 \sqsubseteq \cdots \sqsubseteq x_{N_0+1}$ of non-dummy nodes in the path such that $\mathfrak{s}'(x_i) = \mathfrak{s}'(x_j)$ and $\mathfrak{s}'(x_i\colon x_j) = \mathfrak{s}'(x_{i'}\colon x_{j'})$ for every $0 \le i < j \le N_0 + 1$ and $0 \le i' < j' \le N_0 + 1$. Without loss of generality, we can assume that $\mathfrak{s}'(x_i) < \mathfrak{s}'(x_i\colon x_{i+1})$ for every $0 \le i \le N_0$. Since we only insert separating node prior to a dummy node, we can also assume that all nodes $x_0, x_1, \dots, x_{N_0+1}$ are marked nodes. We have identified a bough backbone $x_0, x_1, \dots, x_{N_0+1}$ of marked nodes defining a bough of level $\mathfrak{s}'(x_0)$, which contradicts the fact that the rewriting procedure terminated. $\square$

$\mathfrak{s}'(x_i) \le \mathfrak{s}'(x_i\colon x_{i+1})$

Proof of Item 3 implies Item 2 of Lemma 4.3. Let us assume that there exists a bound on the dimension of bad boughs, and let us consider a well-quasi-ordered set $(Y, \le)$ of labels. Let $(G_i)_{i\in\mathbb{N}}$ be an infinite sequence of graphs in $\mathsf{Label}_Y(\mathcal{I}(\mathsf{Trees}_M))$. We want to show that there exist two indices $i < j$ such that G_i is a labelled induced subgraph of G_j.

First, every graph G_i can be represented as $\mathcal{I}(\mathcal{T}_i)$ for some tree $\mathcal{T}_i$ equipped with a forward Ramseyan split $\mathfrak{s}_i$. Then, by Lemma 4.4, there exists a finite $L \in \mathbb{N}$ such that we can construct L-bounded well-marked nested tree $(\mathcal{T}_i', \mathfrak{s}_i', \rho_i)$ such that G_i is an induced subgraph of $\mathcal{I}(\mathcal{T}_i')$, when the latter is restricted to vertices corresponding to marked leaves. By labelling the marked leaves of $(\mathcal{T}_i', \mathfrak{s}_i', \rho_i)$ with $\circ$ if they are not in G_i, and with their original label in Y otherwise, we obtain a sequence of $Y \cup \{\circ\}$ node-labelled well-marked nested trees.

By Theorem 3.7, there exist two trees $(\mathcal{T}_i', \mathfrak{s}_i', \rho_i')$ and $(\mathcal{T}_j', \mathfrak{s}_j', \rho_j')$ in this sequence such that $(\mathcal{T}_i, \mathfrak{s}_i, \rho_i) \le_{\text{gap}} (\mathcal{T}_j, \mathfrak{s}_j, \rho_j)$. By Corollary 3.9, the graph interpreted from $(\mathcal{T}_i, \mathfrak{s}_i, \rho_i)$ is a labelled induced subgraph of the graph interpreted from $(\mathcal{T}_j', \mathfrak{s}_j', \rho_j')$, when both are restricted to marked leaves. We conclude that G_i is a labelled induced subgraph of G_j, hence that $\mathsf{Label}_Y(\mathcal{I}(\mathsf{Trees}_M))$ is well-quasi-ordered, and therefore that the image of $\mathcal{I}$ is $\forall$-well-quasi-ordered. $\square$

4.2 Obstructions

In this section, we will prove Item 1 implies Item 3 of Lemma 4.3. We will proceed by contraposition and assume that there exist bad boughs of arbitrarily large dimension, and we will leverage this to construct an infinite antichain in the image of $\mathcal{I}$ using two colours. In order to build such an infinite antichain, we will try to extract from an infinite sequence of bad boughs of increasing dimension some incomparable graphs. To do so, we will first show that we can safely assume that all these bad boughs are compatible (see Lemma 4.5 below), and that one can place all these bad boughs in the *same* context $C[\square]$ to witness that they are bad (see Lemma 4.6 below).

Lemma 4.5. *Let k be a level. There are finitely many boughs of level k up to compatibility.*

Proof. Let B be a bough of level k. We will prove that one can extract finitely many parameters of B, such that any other bough B' of level k sharing these parameters is compatible with B.

To that end, let us fix a context $C[\square]$. Assume that $\mathcal{T} = C[B]$ is forward Ramseyan, and we want to ensure that $\mathcal{T}' = C[B']$

is forward Ramseyan. Recall that $\mathcal{T}'$ is forward Ramseyan if for every branch in $\mathcal{T}'$, for every level l, for every l-neighbourhood in this branch, and for every four nodes $x \sqsubset y$ and $x' \sqsubset y'$ in this l-neighbourhood, we have $[\![x : y]\!]_{\mathcal{T}'} \cdot [\![x' : y']\!]_{\mathcal{T}'} = [\![x : y]\!]_{\mathcal{T}'}$.

First, remark that l-neighbourhoods with $l < k$ inside $\mathcal{T}'$ are fully contained either in B', or $\mathcal{T}_{\text{root}}$, or $\mathcal{T}_{\text{left}}$, or $\mathcal{T}_{\text{right}}$. As a consequence, nothing needs to be checked on such l-neighbourhoods since B' is assumed to be locally forward Ramseyan. More generally, the only possible failures of the forward Ramseyan condition in $\mathcal{T}'$ must involve nodes that are not all contained in one of these four subtrees.

Now, let us consider l-neighbourhoods with $l \geq k$. If $l \geq k$, then the only l-neighbourhood of a branch that is not fully contained in one of the four subtrees crosses B'. Let us distinguish several cases based on the shape of the branch with respect to the context $C[\square]$ and the bough B'.

(1) The branch containing the l-neighbourhood starts in $\mathcal{T}_{\text{root}}$ (possibly in $\square$), goes through B', and ends in $\mathcal{T}_{\text{left}}$ or $\mathcal{T}_{\text{right}}$.
(2) The branch containing the l-neighbourhood starts in $\mathcal{T}_{\text{root}}$ (possibly in $\square$), goes through B', and ends in B'.

In the first case, consider two nodes $x \sqsubset y$ and $x' \sqsubset y'$ in the l-neighbourhood. Remark that the value $[\![x : y]\!]_{\mathcal{T}'}$ can be decomposed into a part before entering B', a part inside B', and a part after leaving B'. By construction, the part inside B' is one of the idempotent elements labelling the backbone of B'. Hence, to ensure that $\mathcal{T}'$ is forward Ramseyan in this case, it suffices to check finitely many equations involving those idempotent elements (note that the number of equations depends on the shape of $C[\square]$, but not the number of elements of M extracted from B').

In the second case, consider two nodes $x \sqsubset y$ and $x' \sqsubset y'$ in the l-neighbourhood. Here, the value $[\![x : y]\!]_{\mathcal{T}'}$ can be decomposed into a part before entering B', a part from b'_0 to the first node z in B' of level l (which is not on the backbone of B' if $l > k$), and a part from z to y. Similarly, the equations ensuring that $\mathcal{T}'$ is forward Ramseyan in this case can be expressed using the values $[\![b'_0 : z]\!]_{\mathcal{T}'}$ for every node z in B', and the values $[\![z : y]\!]_{\mathcal{T}'}$ for every node y in B' in the same l-neighbourhood as z (for some branch). Note that one only cares about the existence of pairs of values $([\![b'_0 : z]\!]_{\mathcal{T}'}, [\![z : y]\!]_{\mathcal{T}'})$ for nodes z and y in the same l-neighbourhood, and there are only finitely many such pairs.

We have proven that, assuming that $\mathcal{T} = C[B]$ is forward Ramseyan, it suffices to check finitely many parameters of B' to ensure that $C[B]$ is forward Ramseyan. Since M is finite, there are finitely many possible values for these parameters, which concludes the proof. $\square$

Lemma 4.6. *There exists a finite set $\mathcal{F}$ of contexts $C[\square]$ such that for every bad bough B of level k, there exists a context $C[\square] \in \mathcal{F}$ such that B is a bad bough in $C[B]$.*

Proof. The proof relies on the fact that the context $C[\square]$ plays a very little role in witnessing that a bough B is bad.

Let B and H be two compatible boughs of level k, and let $C[\square]$ be a context such that $\mathcal{T} = C[B]$ and $\mathcal{T}' = C[H]$ are forward Ramseyan. We want to characterise when there exists an embedding $h \colon \mathcal{I}(\mathcal{T}) \to \mathcal{I}(\mathcal{T}')$ witnessing that B is a good bough in $C[B]$ with respect to H.

First, one only needs to consider the edges between leaves of B or edges between leaves of B and leaves of $C[\square]$. Indeed, the map h acts as the identity on leaves of $C[\square]$, so edges are preserved there.

The fact that there are edges between leaves of B only depends on their respective bough types and the value of $[\![\text{root} : \square]\!]_{\mathcal{T}_{\text{root}}}$. As a consequence, any two contexts sharing the same value for the path from the root to $\square$ do not change the edges between leaves of B (or H). Since there are finitely many monoid elements, there are finitely many possible values for this path.

Let us now consider edges between leaves of B and leaves of $C[\square]$. Similarly as for the leaves in boughs, one can associate to every leaf of $C[\square]$ a *context type* allowing to determine the presence of an edge. For a leaf x in $C[\square]$ that belongs to the root subtree, we collect the following monoid elements: $[\![\text{root} : z]\!]_{\mathcal{T}_{\text{root}}}$, $[\![z : x]\!]_{\mathcal{T}_{\text{root}}}$, and $[\![z : \square]\!]_{\mathcal{T}_{\text{root}}}$, where z is the closest ancestor of x belonging to the path from root to $\square$. For a leaf that belongs to the left (resp. right) subtree, we only consider the monoid element $m_{\text{left}} \cdot [\![\text{root} : x]\!]_{\mathcal{T}_{\text{left}}}$ (resp. $m_{\text{right}} \cdot [\![\text{root} : x]\!]_{\mathcal{T}_{\text{right}}}$). It is immediate that to compute the presence of an edge between a leaf y in B and a leaf x in $C[\square]$, it suffices to know the bough type of y and the context type of x (see Equation (1), and Figure 5). Since there are finitely many possible context types and bough types, if a bough B is a good bough in $C[B]$ using a map h and a compatible bough H, then for any other context $C'[\square]$ sharing the same context types for its leaves as $C[\square]$, the bough B is also a good bough in $C'[B]$ using the same map h and the same compatible bough H.

Since there are finitely many possible contexts up to this equivalence relation, we conclude the proof. $\square$

Proof of Item 1 implies Item 3 of Lemma 4.3. Let us assume towards a contradiction that the class of graphs obtained as images of $\mathcal{I}$ is 2-well-quasi-ordered, but that there exist bad boughs of arbitrarily large dimension. Let $(B_i)_{i \geq 1}$ be an infinite sequence of such bad boughs. Without loss of generality, one can assume that all boughs B_i have the same level k, and are pairwise compatible, by Lemma 4.5.

Using Lemma 4.6, one can extract one context $C[\square]$ such that infinitely many boughs B_i are bad boughs in $C[B_i]$. Finally, one can extract further to assume that the dimension of the boughs B_{i+1} is greater than thrice the number of leaves in B_i.

Let us now colour every graph $\mathcal{I}(C[B_i])$ with two labels: $\top$ to leaves belonging to B_i, and colour $\bot$ to leaves belonging to $C[\square]$. Because we assume that the image of $\mathcal{I}$ is 2-well-quasi-ordered, we can extract an infinite increasing subsequence, and assume that for all $i < j$, there exists an embedding $f_{i,j} \colon \mathcal{I}(C[B_i]) \to \mathcal{I}(C[B_j])$ that preserves labels. Let us remark that for all $i < j$, the map $f_{i,j}$ acts as a permutation when restricted to nodes labelled $\bot$, that is of size the number of leaves in $C[\square]$. By a Ramsey argument, we can therefore assume that this permutation is the identity for all $i < j$.

Finally, consider any two indices $i < j$. Because the dimension of B_j is greater than thrice the number of leaves in B_i, there must exist three consecutive bough blocks of B_j that are left untouched by $f_{i,j}$. Furthermore, we know that $f_{i,j}$ is the identity on leaves belonging to $C[\square]$. Hence, the map $f_{i,j}$ witnesses that B_i is a good bough in $C[B_i]$, which is a contradiction. $\square$

5 Bounded Clique Width

In this section, we will aim at leveraging the previous results to tackle classes of bounded clique-width. Our first goal is to prove our Theorem 1.5 regarding the images of finite trees under MSO-interpretations. There are two difficulties here: first, we only dealt with simple MSO-interpretations so far, and second, we have not yet given any decision procedure to check whether the image of a given simple MSO-interpretation is 2-well-quasi-ordered. Let us first tackle the second difficulty.

Definition 5.1 (Perfect Bough). Let $\mathcal{T} = C[B]$ be a tree with a bough B of level k. We say that B is a *perfect bough* there exists a compatible bough H of level k and a map $h\colon \mathcal{I}(C[B]) \to \mathcal{I}(C[H])$

(1) h is an embedding of graphs,
(2) h is the identity map on leaves belonging $C[\square]$
(3) the bough type of every leaf x in H is the same as the bough type of $h(x)$ in B.
(4) there exist three consecutive blocks in H that are left untouched by h.

An *imperfect bough* is a bough that is not a perfect bough.

Remark that every perfect bough is a good bough, hence every bad bough is an imperfect bough. This allows us to lift Lemma 4.3 to perfect boughs.

Lemma 5.2. *The image of $\mathcal{I}$ is 2-well-quasi-ordered if and only if the image of $\mathcal{I}$ is $\forall$-well-quasi-ordered if and only if there exists a bound on the dimension of imperfect boughs.*

Proof. First, if we have a bound on the dimension of imperfect boughs, then we also have a bound on the dimension of bad boughs, because every bad bough is an imperfect bough. Hence, by Lemma 4.3, the image of $\mathcal{I}$ is $\forall$-well-quasi-ordered.

Conversely, assume that there exist imperfect boughs of arbitrarily large dimension. Then, by an argument similar to the one of Section 4.2, we can construct an infinite labelled antichain in the image of $\mathcal{I}$. The only change is that one also adds the bough type of the leaves as additional labels. This actually follows the original proof technique of [18] for bounded linear clique-width. $\square$

The important advantage of Lemma 5.2 over Lemma 4.3 is that imperfect boughs are easily recognizable using Monadic Second-Order logic on trees. Indeed, every bough B is actually a (binary) tree with some distinguished nodes that form its backbone. Hence, given a tree $\mathcal{T}$ together with a split $\mathfrak{s}$ and labels on its edges coming from a finite monoid M, one can ask whether this tree with distinguished nodes represents an imperfect bough.

In order to do so, we will first need the following Fact 5.3, that explains that when replacing a bough B by a compatible bough H, one does not change the part of the graph represented by the leaves of the bough context $C[\square]$. This is because to determine the presence of edges between leaves of $C[\square]$, one only needs to know the value of the first idempotent element labelling the bough, which does not change when replacing B by a compatible bough H. As a consequence, it suffices to focus on the leaves of B inside of H to construct an embedding from $\mathcal{I}(C[B])$ to $\mathcal{I}(C[H])$ satisfying the conditions of Definition 5.1.

Fact 5.3. Let $\mathcal{T} = C[B]$ be a tree with a bough B of level k, and let H be a bough that is compatible with B. Let $\mathcal{T}' = C[H]$ be the tree obtained by replacing B by H in $\mathcal{T}$. Then, the subraph of $\mathcal{I}(\mathcal{T})$ induced by the leaves in the context $C[\square]$ equals the subgraph of $\mathcal{I}(\mathcal{T}')$ induced by the leaves in the context $C[\square]$.

Lemma 5.4. *There exists an* MSO *formula that recognizes imperfect boughs.*

Proof. The proof follows the same pattern as [18, Lemma 19], but adapts it from words to trees. Assume that B is a perfect bough of level k inside of a tree $\mathcal{T} = C[B]$. This gives us a compatible bough H of level k and an embedding $h\colon \mathcal{I}(C[B]) \to \mathcal{I}(C[H])$ satisfying the conditions of Definition 5.1.

Let us consider five copies of B, called $B_{\mathsf{left}}, B_{\mathsf{mid}-1}, B_{\mathsf{mid}-2}, B_{\mathsf{mid}-3}$, and B_{right}, and let us define B^5 to be the bough obtained by concatenating these five copies merging b_n of B_{left} with b_0 of $B_{\mathsf{mid}-1}$, and merging b_n of $B_{\mathsf{mid}-1}$ with b_0 of $B_{\mathsf{mid}-2}$ and so on. We claim that B and B^5 are compatible boughs.

Let us define a mapping $k\colon \mathcal{I}(C[B]) \to \mathcal{I}(C[B^5])$ as follows: leaves x of B are mapped to their corresponding leaf in B_{left} or B_{right} depending on whether $h(x)$ is to the left or right of the untouched bough blocks of H. For leaves x of $C[\square]$, we let $k(x) = x$. Thanks to Fact 5.3, we know that k preserves edges between leaves of $C[\square]$. Furthermore, because h is an embedding that preserves the bough type of leaves in B, and because bough types are enough to compute the presence of an edge between a leaf of B and a leaf of $C[\square]$ (or between two leaves of B), we deduce that k is also an embedding.

What we have shown is that if B is a perfect bough, then there exists a mapping k' from leaves of B to $\{\mathsf{left}, \mathsf{right}\}$ such that there is an edge between two leaves x and y of B in the graph $\mathcal{I}(C[B])$ if and only if the bough types of x and y are such that there is an edge between $x_{k'(x)}$ and $y_{k'(y)}$ in the graph $\mathcal{I}(C[B^5])$. One can guess such a mapping k' in Monadic Second-Order logic, and then verify the required conditions on edges between leaves of B. As a consequence, one can write an MSO formula that holds if and only if B is a perfect bough. $\square$

Corollary 5.5. *One can decide whether there are imperfect boughs of arbitrarily large dimension.*

Proof. We will use the theory of Cost-MSO formulas [6, Section 6]. Let us call Φ the formula of Lemma 5.4 that recognizes imperfect boughs. Let us now consider the following Cost MSO formula:

$$\Psi(N) \triangleq \Phi \wedge |\{\text{ dimension of the bough }\}| \geq N$$

This formula takes an input parameter N (a natural number), and decides whether the provided tree with split and labels is an imperfect bough of dimension at most N. Because one can compare the boundedness of Cost-MSO definable functions [6, Corollary 16], one can decide whether such a formula can hold for arbitrarily large values of N, effectively answering our question. $\square$

We are now ready to prove our main Theorem 1.5.

Theorem 1.5. *Let $\mathcal{C}$ be the image of a class of finite trees under an* MSO *interpretation $\mathcal{I}$. Then, the following are equivalent and decidable given $\mathcal{I}$:*

(1) *$\mathcal{C}$ is 2-well-quasi-ordered,*
(2) *$\mathcal{C}$ is $\forall$-well-quasi-ordered,*

(3) *$\mathcal{C}$ is orderably $\forall$-well-quasi-ordered,*

Proof. Let $\mathcal{I}$ be an MSO-interpretation. We can get rid of the domain formula and selection formula by standard arguments without changing the 2-well-quasi-ordering status of the image of $\mathcal{I}$ [18, Lemma 8]. Hence, we can assume that $\mathcal{I}$ is a simple MSO-interpretation. Furthermore, the transformation from an simple MSO-interpretation to a monoid interpretation from trees described in Section 3 is effective. Finally, by Lemma 5.2 the image of $\mathcal{I}$ is 2-well-quasi-ordered if and only if it is $\forall$-well-quasi-ordered, and by Lemma 5.4, one can decide whether the image of $\mathcal{I}$ is 2-well-quasi-ordered.

It remains to discuss the fact that one can add a total ordering on the graphs without changing the 2-well-quasi-ordering status of the image of $\mathcal{I}$. This is because we prove that the image of $\mathcal{I}$ is 2-well-quasi-ordered using the marked gap-embedding ordering on marked nested trees representing the graphs, which already includes a total ordering on the leaves. □

It is straightforward to derive that the same holds for hereditary classes of bounded clique-width, thus proving Corollary 1.6.

Proof. By a standard argument, if the class $\mathcal{I}$ is hereditary and 2-well-quasi-ordered, then it is characterised by finitely many forbidden induced subgraphs [9, Proposition 3]. Hence, if $\mathcal{C}$ has bounded clique-width and is hereditary, one can assume that $\mathcal{C}$ is exactly the image of an MSO-interpretation $\mathcal{I}$ from finite trees: this is done by ensuring in the formula ϕ_{dom} that the graph that would be produced does not contain any of these forbidden induced subgraphs. The result then follows from Theorem 1.5. □

Let us now turn our attention to Theorem 1.7, and in particular the link between 2-well-quasi-ordering and bounded linear clique-width. The following lemma shows that the existence of imperfect boughs of arbitrarily large dimension can be witnessed using only blocks from a finite set.

Lemma 5.6. *Assume that there are imperfect boughs of arbitrarily large dimension. Then, there exists a finite set of blocks $\mathbb{F}$ such that one can build an imperfect bough of arbitrarily large dimension using only blocks from $\mathbb{F}$.*

Proof. The proof follows the same pattern as Lemmas 4.5 and 4.6, and continues on the ideas of Lemma 5.4. The idea is that if a bough B is perfect, then one can witness it by using the compatible bough B^5 defined by gluing five copies of B together, because we noticed that perfectness essentially depends on the bough types of the leaves of B. Another consequence of this analysis is that if one defines an equivalence relation between bough blocks of a given bough B based on the set of available bough types of its leaves, then one can always replace a bough block by another one in the same equivalence class without changing the perfectness of B. We conclude because this equivalence relation has finitely many classes. □

As an immediate consequence of Lemma 5.2 and Lemma 5.6, we can transform long imperfect boughs into long imperfect boughs that are using only blocks from a finite set. This allows us to prove the following corollary.

Corollary 5.7. *Assume that the class $\mathcal{I}$ is hereditary. Then, the image of $\mathcal{I}$ is 2-well-quasi-ordered if and only if every subclass of bounded linear clique-width in $\mathcal{I}$ is 2-well-quasi-ordered.*

Proof. If the image of $\mathcal{I}$ is 2-well-quasi-ordered, then every subclass of bounded linear clique-width in $\mathcal{I}$ is also 2-well-quasi-ordered. On the other hand, if the image of $\mathcal{I}$ is not 2-well-quasi-ordered, then by Lemma 5.2 there exist imperfect boughs of arbitrarily large dimension. Let us consider the finite set of blocks $\mathbb{F}$ given by Lemma 5.6, to obtain an infinite sequence of imperfect boughs $(B_i)_{i\in\mathbb{N}}$ of increasing dimension using only blocks from $\mathbb{F}$.

Note that if B is an imperfect bough in some tree $\mathcal{T} = C[B]$, then it remains an imperfect bough in any context $C'[\square]$, as the definition of perfect bough is local to the bough itself. Hence, one can select a single context $C[\square]$ such that B_i is an imperfect bough inside of $C[\square]$ for all $i \in \mathbb{N}$. Similarly as in Section 4.2, one can then build an infinite antichain of two-labelled graphs inside of $\mathcal{I}$ by colouring vertices of $\mathcal{I}(C[B_i])$ by $\circ$ if they come from $C[\square]$ and by $\bullet$ if they come from B_i.

Finally, remark that the collection of graphs $\mathcal{I}(C[B_i])$ for $i \in \mathbb{N}$ has bounded linear clique-width. Indeed, the context $C[\square]$ is fixed, and the blocks used to build the boughs B_i come from a finite set $\mathbb{F}$. As a consequence, one can represent these graphs by using one colour per node of the context $C[\square]$ plus one colour per node in each block in $\mathbb{F}$, and write a single branch (linear) expressions representing all graphs $\mathcal{I}(C[B_i])$. □

6 Structural Characterisations

In this section, we will be interested in extracting a very regular structure from a graph class $\mathcal{C}$ that is not $\forall$-well-quasi-ordered. This structure will take two forms, the first one is a combinatorial notion of periodic antichain that is similar to the one proposed in [4, Section 7], and the second one is a logical property of existentially transducing the class of all finite paths.

Definition 6.1 (Periodic Sequence). A *periodic sequence (of graphs)* is given by a finite graph G, a labelling function $\ell\colon V(G) \to \Sigma$ for some finite set Σ, and two sets $C, F \subseteq \Sigma^2$. It defines an infinite set of graphs $(G^r)_{r\geq 1}$ as follows: the vertex set of G^r is $V(G) \times \{1, \dots, r\}$, and there is an edge between (u, i) and (v, j) with $i \leq j$ if and only if one of the following holds:

- $(u, v) \in E(G)$ and $i = j$;
- $(\ell(u), \ell(v)) \in C$ and $|i - j| = 1$
- $(\ell(u), \ell(v)) \in F$ and $|i - j| > 1$.

It follows from results of the previous section that one can extract periodic antichains from any hereditary class of graphs that is not $\forall$-well-quasi-ordered.

Lemma 6.2. *If a hereditary class of graphs $\mathcal{C}$ is not $\forall$-well-quasi-ordered, then it contains a periodic sequence $(G^r)_{r\geq 1}$ that is an antichain for the labelled induced subgraph relation.*

Proof. Using Corollary 5.7, one can assume that $\mathcal{C}$ has bounded linear clique-width. It then follows from the analysis of [18] that $\mathcal{C}$ contains a periodic sequence $(G^r)_{r\geq 1}$. □

Let us now leverage this combinatorial structure to extract a logical property of the class $\mathcal{C}$.

Lemma 6.3. *Let C be a periodic antichain (having the coloring defined) as in Lemma 6.2. Then C existentially interprets the class of all finite paths.*

Proof.

Mael: Draft, most arguments are here, let me know if there is something that sounds fishy, i'm not used to write such things. I will rewrite it nicer later

Let $C = (G^r)_{r\geq 1}$ be a periodic antichain with labelling function $\ell\colon V(G) \to \Sigma$ and sets $C, F \subseteq \Sigma^2$. Moreover assume that ℓ is injective up to duplicating labels from Σ, in particular $|\Sigma| = |V(G)|$. For $i \leq r$, we denote the i-th copy of G in G^r by G_i, and by $u_i \in G^r$ the i-th copy of $u \in V(G)$. We will proceed in two steps, we first define a formula producing a directed graph where there is a directed shortest path from G_1 to G_r going through every G_i, but where there is may be arcs going from G_i to G_j for $j \leq i$. We will then show how to get rid of theses back arcs, and produce the path P_r.

We now define formula:

$$\phi_{\to}(u, v) = \neg(uv \in E(G^r) \Leftrightarrow l(u)l(v) \in F)$$

Informally, $\phi_{\to}(u, v)$ creates an arc (u, v) if the edge relation between $u, v \in G^r$ differs if u and v are put far away form each other in this order (note that u can be in a copy of G that comes after the one of v in G^r). Let $H^r = \mathcal{I}(G^r)$ for $\mathcal{I} = (\top, \phi_{\to}, \top)$, note that H is a directed graph as $\phi_{\to}$ is not symmetric.

Claim 6.4. For $u_iv_j \in E(H^r)$ only if $j \leq i + 1$.

Proof. If $j \geq i + 1$, then $uv \in E(G^r)$ if and only if $l(u)l(v) \in F$ by definition of G^r. ■

Claim 6.5. If $u_iv_j \in E(H^r)$ for $j < i$ then for every $j' \leq i$, $u_iv_{j'} \in E(H^r)$.

Proof. Follow from the definiton. ■

A spanning path of H^r is a directed shortest path from $V(H_1)$ to $V(H_r)$ going through at least one vertex from each $V(H_i)$.

Claim 6.6. There is a spanning path in H^r.

Proof. Let V_i^+ be the vertices of H_i such that there is an arc from H_{i-1} to them, V_i^- be the vertices of H_i such that there is an arc from them to H_{i+1}, and V_i^o be the remaining vertices of V_i. Suppose there is no spanning path in H^r, Claim 6.4, it have to be because either there is no path in H_i from V_i^+ to V_i^-, or that they are empty (i.e. there is no arc from H_i to H_{i+1}).

Mael: Argue that in this case we can embed G^r into G^{r+1}, contradicting the antichain.

■

Let P a spanning path of H^r, furthermore assume that it is periodic, i.e. that P go through the same sequence of vertices in each H_i, say $P_i = u_i^1 \dots u_i^p$, and $P = P_1P_2 \dots P_r$. If there is a backward arc $uv \in E(H^r)$ in P with $u \in P_i$ and $v \in P_j$ with $i > j$, then by Claim 6.5 this arc exists between every copy of u and v where v is an earlier copy than the one of u. Define now the formula

$$\phi_P(u, v) = \mathrm{dist}_P(u, v) \leq p \wedge \mathrm{dist}_P(v, u) \leq p \wedge l(u) = l(v) = c,$$

where $c = \ell(u_1^p)$. This formula create a path from u_1^p to u_{r-1}^p, moreover this path is undirected as $\phi_P(u, v)$ is symmetric. Consider the interpretation $\mathcal{I} = (\top, \phi_P, \phi_{univ})$, where

$$\phi_{univ}(u) = l(u) = c \wedge \exists v, \phi_P(u, v)$$

(the exist is to remove the vertex u_r^p). It follows that $\mathcal{I}((G^r)_{r\geq 2})$ is the class of all path. It is easy to see that these formula are existential first order.

□

Corollary 6.7. *If a hereditary class of graphs C is not $\forall$-well-quasi-ordered, then it existentially transduces the class of all finite paths.*

of bnd cw?

Proof.

MAEL ICI

It follows from Lemma 6.2 and Lemma 6.3. □

Mael: Proof of Corollary 6.7

6.1 From Transductions to Interpretations

Let us recall the definition of weak periodic antichain from [4, Conjecture 2]: given a finite alphabet Σ, and a finite set $S \subseteq \Sigma^2$, one can map a word $w \in \Sigma^*$ to a graph G_w whose vertex set is the positions of w, and where there is an edge between two positions $i + 1 < j$ if and only if $(w[i], w[j]) \in S$, and an edge between two consecutive positions i and $i + 1$ if and only if $(w[i], w[i + 1]) \notin S$. In this setting, a *(weak) periodic sequence* is given by a finite word $u \in \Sigma^+$, and a finite set $S \subseteq \Sigma^2$, and defines the infinite set of graphs $(G_{u^r})_{r\geq 1}$. It is easy to see that every weak periodic sequence is also a periodic sequence as defined in this paper, by grouping the vertices of u into a single graph G and adapting colors. Finally, a *weak periodic antichain* is a weak periodic sequence that is an antichain for the labelled induced subgraph relation when the first and last letters of the words u^r are given distinct colors.

We now prove that [4, Conjecture 2] holds for hereditary classes of graphs having bounded clique-width.

Lemma 6.8. *If a hereditary class of graphs C is not $\forall$-well-quasi-ordered, then it contains a weak periodic antichain.*

Proof. It follows from Lemma 6.2 and the fact that every periodic sequence contains a weak periodic sequence by coloring the first and last vertices of G with distinct colors. Indeed, this is what we did in the proof of Lemma 6.3. □

From this, we can now try to prove the following conjecture, which would complete the structural characterization of hereditary classes of graphs that are $\forall$-well-quasi-ordered.

Conjecture 6.9. *A hereditary class of graphs C is $\forall$-well-quasi-ordered if and only if it does not existentially interprets the class of all finite paths.*

Sketch of proof. The only difficult direction is to actually interpret paths, not using colors to identify the behaviour of the nodes inside the periodic sequence.

TODO: maybe it is possible to adapt the proof!

□

7 Concluding Remarks

We have proven that for all classes of graphs of bounded clique-width, the wqo-wqo and labelled-wqo are two equivalent notions. This result is a first step towards proving this conjecture for arbitrary classes of graphs.

- A natural generalisation is to consider classes of finite relational structures that are of bounded clique-width (images of finite labelled trees under MSO interpretations). There are no main obstacles to extending our techniques, except that the *accepting parts* defining monoid interpretations become more complex to handle
- What about classes that are monadically dependent ?
- The only ordering that exists on graphs that is WQO is when they are looking like trees...

Acknowledgments

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Received 20 February 2007; revised 12 March 2009; accepted 5 June 2009